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1. A mobile app enhances photos automatically. Identify whether this task belongs to image processing or computer vision and justify.

Answer: Image Processing

Explanation: It only modifies the pixels to improve visual quality like brightness and noise removal. It does not understand or recognize what is inside the image, so it is not computer vision.

2. A game engine generates a 3D virtual environment. Classify this under computer graphics or vision and explain.

Answer: Computer Graphics

Explanation: Because it creates synthetic images from 3D models. Graphics is Image Synthesis, while Vision is Image Analysis.

3. A system detects edges in an image before object recognition. Which level of vision is involved?

Answer: Low-level vision

Explanation: Low-level vision extracts basic primitive features like edges, corners, color and texture without understanding the meaning of the image.

4. A robot identifies objects after extracting features. Which vision level is this?

Answer: High-level vision

Explanation: Identifying and recognizing objects by using extracted features requires interpretation and understanding of the scene, which is high-level vision.

5. A CCTV system identifies a person from a crowd. Which vision level is involved?

Answer: High-level vision

Explanation: Person identification involves face/body recognition and scene understanding, which comes under high-level vision.

6. A camera captures a blurred image due to improper focus. What aspect of image formation is affected?

Answer: Optical / Focus aspect is affected

Explanation: Due to improper focus, light rays do not converge correctly on the sensor plane. This causes defocus blur and is related to focal length and depth of field.

7. A smartphone camera shows distorted images. What is the need for camera calibration here?

Answer: To correct lens distortion

Explanation: Calibration is needed to find intrinsic parameters like focal length, principal point, and distortion coefficients to correct radial and tangential distortion.

8. A self-driving car uses two cameras to estimate distance. What concept is being used?

Answer: Stereo Vision / Stereopsis

Explanation: Depth is calculated from the disparity between two images taken from two slightly different viewpoints, similar to human eyes.

9. A drone captures images from different angles to build a 3D model. Which concept is applied?

Answer: Structure from Motion (SfM)

Explanation: 3D structure of the scene is reconstructed from multiple 2D images taken from different viewpoints.

10. A medical imaging system analyzes MRI scans to detect tumors. Which CV application area does this belong to?

Answer: Medical Image Analysis

Explanation: It is a biomedical application domain of computer vision used for diagnosis and analysis of medical scans.

11. A virtual try-on app overlays glasses on a user's face in real time. Which fields are involved: graphics, vision, or both?

Answer: Both Graphics and Vision

Explanation: Vision is used to detect face and track facial landmarks. Graphics is used to render and overlay the 3D model of glasses on the face.

12. A system converts a real-world 3D scene into a 2D image using a camera. What process of image formation is taking place?

Answer: Perspective Projection

Explanation: The 3D world points are projected onto the 2D image plane through the camera lens using perspective transformation.

13. A smartphone camera app corrects geometric distortion in captured images. Which concept is applied here?

Answer: Camera Calibration and Distortion Correction

Explanation: The system applies radial and tangential distortion correction using calibration parameters to make the image geometrically correct.

14. A surveillance system detects motion but cannot identify objects. Which level of vision is represented?

Answer: Low-level to Mid-level vision

Explanation: It can only detect changes and movement but cannot recognize or understand what the moving object is, so it has not reached high-level vision.

15. A robot identifies objects after analyzing their shape and features. Which level of vision is involved?

Answer: High-level vision

Explanation: Analyzing shape is mid-level, but using that shape to identify the object is recognition, which is high-level.

16. A VR system creates a virtual environment without using real-world images. Is this computer graphics or vision? Justify.

Answer: Computer Graphics

Explanation: It is completely synthesized from 3D models without analyzing any real images. So it is pure graphics, not vision.

17. A stereo camera system generates depth maps for a scene. Which concept is being applied?

Answer: Stereo Vision / Depth from Disparity

Explanation: Disparity between left and right images is calculated and converted into depth information to create a depth map using triangulation.

18. A system analyzes multiple images of a building to estimate its structure. Which concept is used here?

Answer: Structure from Motion / 3D Reconstruction / Photogrammetry

Explanation: The 3D shape and structure of the building is estimated from multiple 2D views from different angles.

19. A biometric system uses facial features to verify identity. Which application domain of computer vision does this belong to?

Answer: Biometrics / Face Recognition

Explanation: It belongs to the biometric identification and pattern recognition application domain of computer vision.

20. A camera fails to measure accurate depth due to improper alignment of lenses. Which concept needs correction?

Answer: Stereo Calibration and Rectification

Explanation: Extrinsic parameters like rotation and translation between the two cameras need to be corrected to align epipolar lines and get accurate depth.